AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271
Machine Learning and Deep Learning

Final Examination
Summer Semester 2024/2025

Date: 6 /Sep./2025

Number of Exam Pages: (3) (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **NOT** allowed.

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PART – ONE Total Marks 10

Question No 1: Select the correct choice for each of the following statements.

1. **Which activation function is commonly used in CNNs for handling non-linearity?**

A) Linear

B) Sigmoid

C) ReLU (Rectified Linear Unit)

D) Tanh

2. **Recurrent Neural Networks (RNNs) excel at tasks involving:**

A) Recognizing Audio

B) Predicting time series

C) Recognizing speech

D) All of the above

3. **In the context of Natural Language Processing (NLP), tokenization serves to:**

A) Extract features

B) Reduce dimensionality

C) Segment text into words or sub words

D) Shorten the length of sequences

4. **The role of the loss function in image classification is _____**

A) Measuring the model's performance

B) Generating synthetic images

C) Reducing image resolution

D) Controlling the learning rate

5. **What is the purpose of the collaborative filtering technique in recommender systems?**

A) Analyzing user demographics

B) Recommending items based on user-item interactions

C) Extracting features from item descriptions

D) Ranking items based on popularity

PART – TWO Total Marks 15

Question No 2: Compare the architectural design of GoogleNet and ResNet models **(6 Marks)**

Question No 3: Give a short explanation of your approach to building a time series model that predicts multiple future time steps. **(5 Marks)**

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Question No 4: What is a recommender system, and where can it be applied? Give two examples. **(4 Marks)**

PART – THREE (Total Marks 25)

For the following questions, you are required to write the python code.

Scenario: Energy Demand Forecasting Using RNNs

You are a data scientist at a smart energy solutions company. Your team has collected a dataset called **"energy_consumption.csv"**, which contains hourly electricity usage data from residential and commercial sectors over the past 10 years. The dataset includes 200,000 rows and also features related variables such as weather conditions, public holidays, and economic activity indexes. Your task is to design and train a **Recurrent Neural Network (RNN)** to predict **future electricity demand**. The model should capture temporal dependencies and adjust for factors like seasonal trends and weather changes.

Question No 1: Loading the Dataset **(3 Marks)**

Write the Python code to load the dataset "energy_consumption.csv" into a DataFrame named "enery_consumption".

Question No 2: Dataset Splitting **(5 Marks)**

Write the Python code to divide the "enery_consumption" DataFrame into training and testing datasets. Assume that there are 200,000 rows in the "enery_consumption".

Question No 3: RNN Architecture Design **(12 Marks)**

Design an RNN architecture with at least 3 layers for forecasting energy consumption trends. Explain your choice of layers, activation functions, and any other architectural decisions. Feel free to modify the number of layers as necessary.

```
from keras.models import Sequential
from keras.layers import SimpleRNN, Dense, Dropout
model = Sequential()
##Write your code after here
```

Question No 4: Model Training and Prediction **(5 marks)**

Write a Python code to train the designed RNN model using the training dataset. Also, provide code for making predictions on the test dataset.

End of questions

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